

2016.0 RANGE ROVER (LG), 205-02

REAR DRIVE AXLE/DIFFERENTIAL

DRIVE PINION SEAL – TDV6
3.0L DIESEL /TDV8 4.4L
DIESEL/V6 S/C 3.0L PETROL
/V8 N/A 5.0L PETROL/V8
S/C 5.0L PETROL (G1509002)

REMOVAL AND INSTALLATION

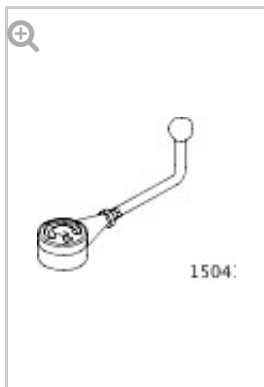
			3000 CC, TDV6, DIESEL, WITH PARTICULATE FILTER, WITH DIESEL EXHAUST FLUID			
51.20.02	OIL SEAL - PINION - DIFFERENTIAL HOUSING - RENEW			4.8	USED WITHINS	+

SPECIAL TOOL(S)



100-012

Slide Hammer



205-067

Preload Gauge



303-1300

Remover,
Crankshaft Seal



E164035

307-499

Installer, Output
Shaft Inner Seal



E52536

307-520

Installer, Output
Shaft Seal



E164032

JLR-205- 1011

Remover, Bearing



E164033

JLR-205- 1012

Installer, Bearing



E164028

JLR-205-890

Support,
Differential



E164029

JLR-205-890-1

Pinion Support,
Differential Case



E145956

JLR-205-994

Holder, Rear
Differential Pinion



E145958

JLR-205-996

Remover/Installer,
Differential Pinion
Nut



E145964

JLR-205-998

Installer, Rear
Differential Seal

PART(S)

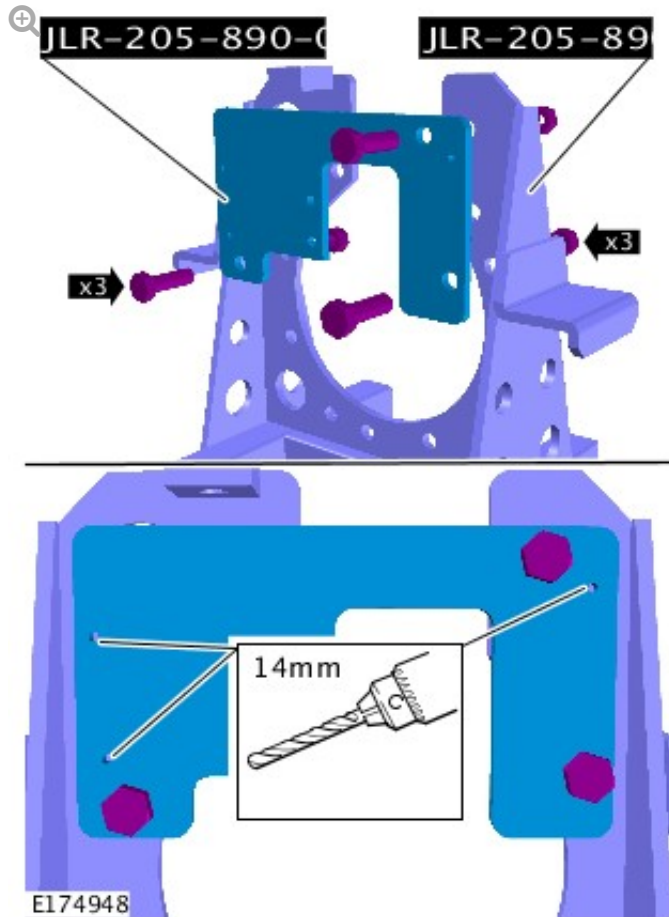
STEP	REPLACE PART / RENEW PART	PART NAME
Installation Step 1	Renew Part	Differential pinion seal
Installation Step 2	Renew Part	Rear differential pinion repair kit
Installation Step 15	Renew Part	Rear differential pinion repair kit
Installation Step 15	Renew Part	Circlip

REMOVAL

CAUTION:

Torque to rotate should always be measured and recorded at step 9 before removing the drive pinion nut.

1.



Special Tool(s): [JLR-205-890](#), [JLR-205-890-1](#)

2. Refer to: [Differential Draining and Filling](#) (205-02 Rear Drive Axle/Differential, General Procedures).
3. Refer to: [Differential Case - TDV6 3.0L Diesel /TDV8 4.4L Diesel/V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol](#) (205-02 Rear Drive Axle/Differential, Removal and Installation).

4.

NOTE:

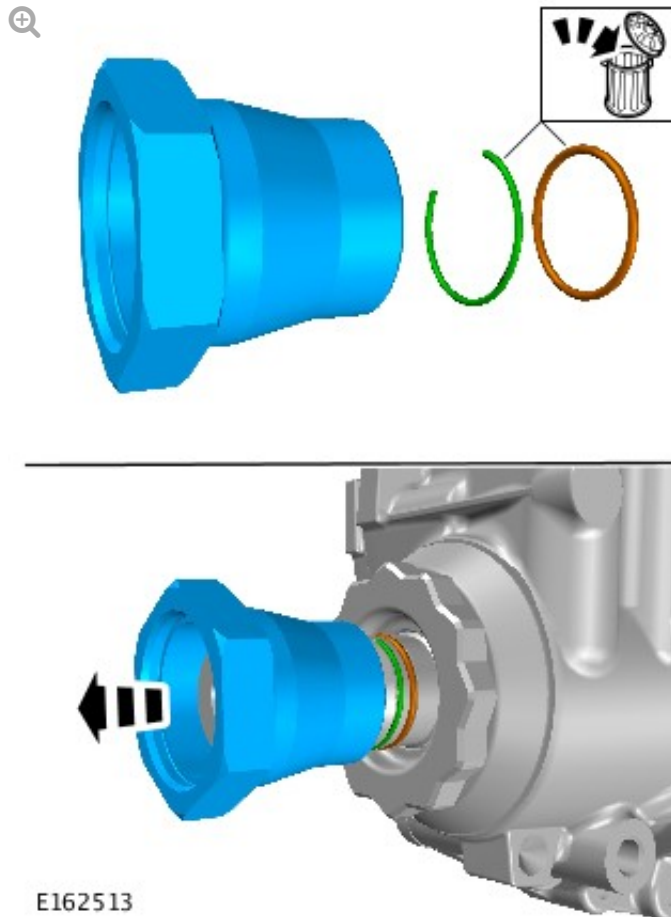
If equipped.

Refer to: [Differential Locking Motor](#) (205-02 Rear Drive Axle/Differential, Removal and Installation).

5.

NOTES:

- Remove and discard the drive pinion circlip.
- Remove and discard the drive pinion o-ring seals.

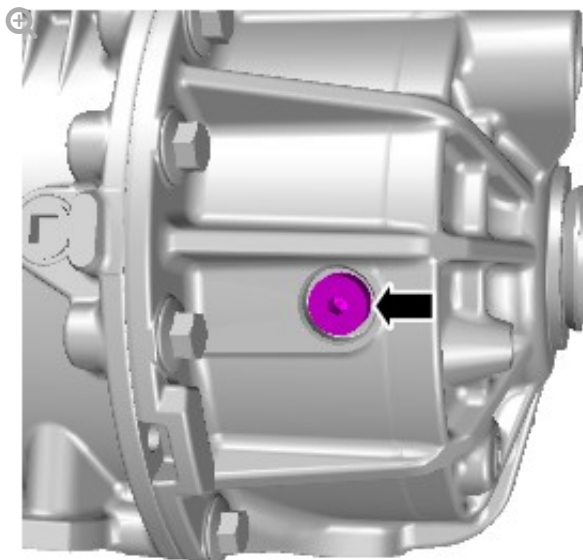


Remove the drive shaft nut.

6.

NOTE:

If equipped.



E194640

Remove the electronic differential locking pin.

7.

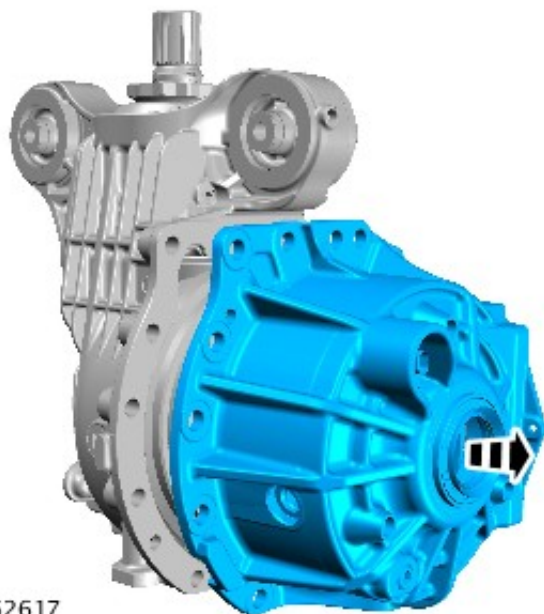


E162616

8.

CAUTION:

Be prepared to collect escaping oil.



E162617

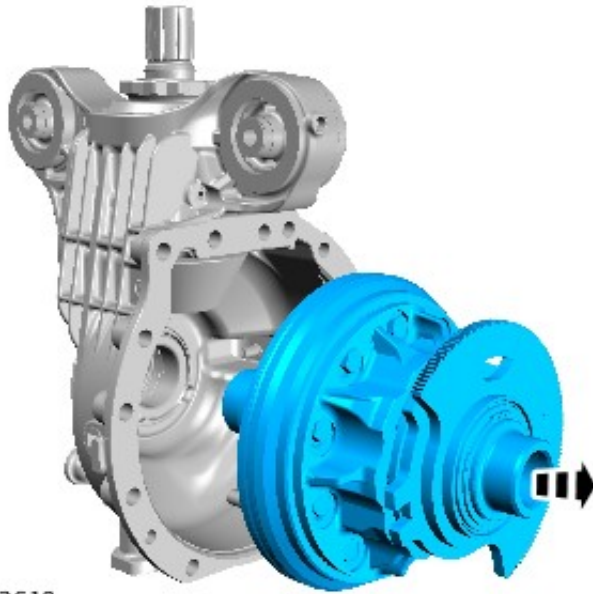
9.

WARNING:

Remove sealant from casing threads and bolts.

NOTE:

Clean all the mating faces thoroughly and check for damage.

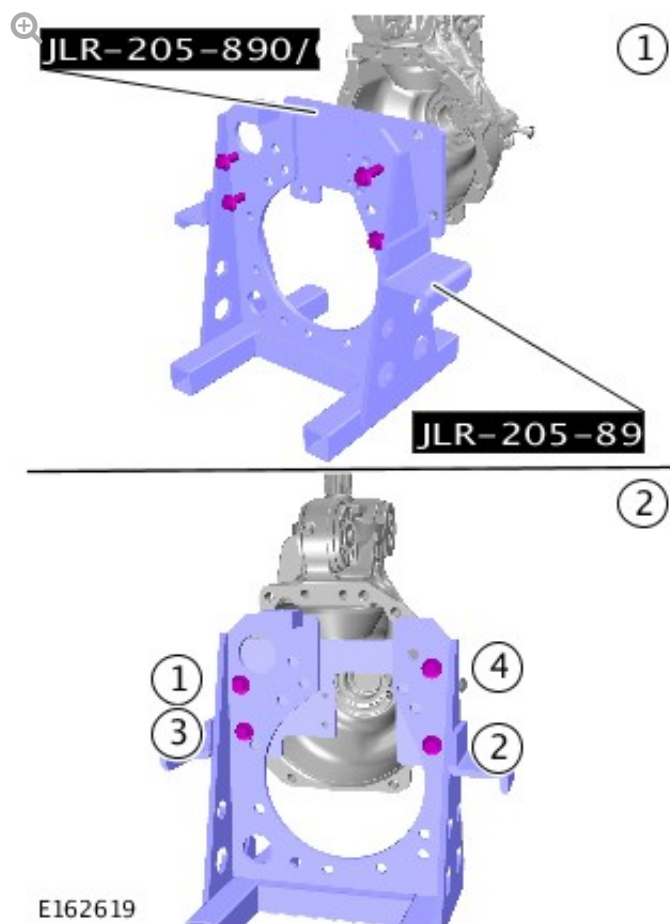


E162618

10.

NOTE:

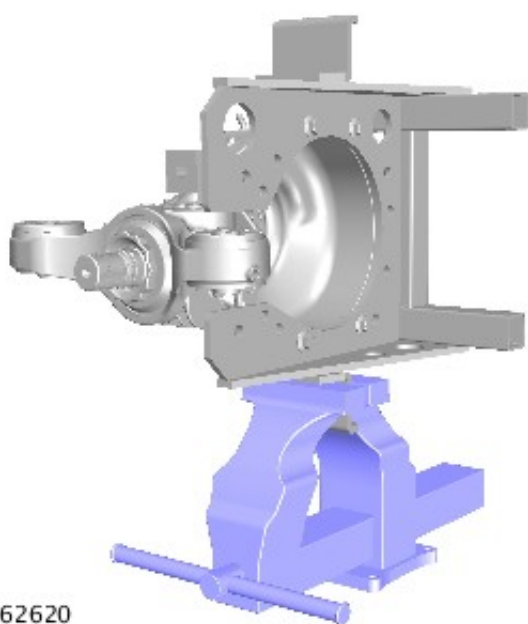
Tighten the retaining bolts in the sequence illustrated.



Special Tool(s): [JLR-205-890](#), [JLR-205-890-1](#)

Torque: 45 Nm

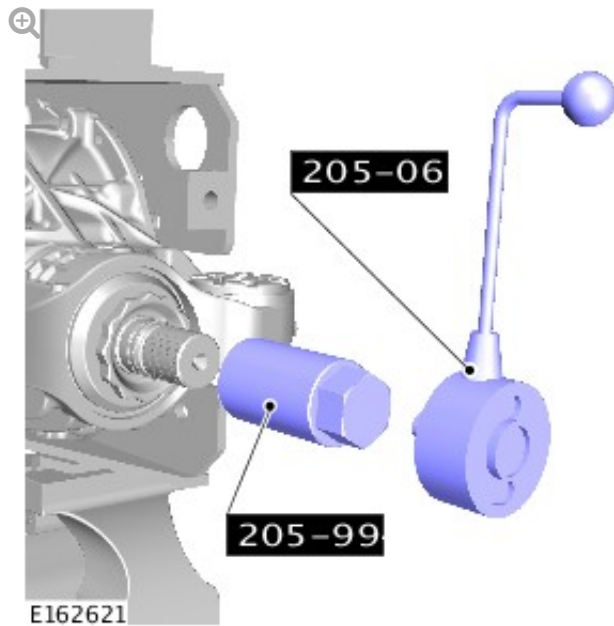
11.



12.

NOTES:

- Check and record torque to rotate before removing the drive pinion assembly any further.
- The special tool 205-067 should be rotated clockwise to check the torque to turn.

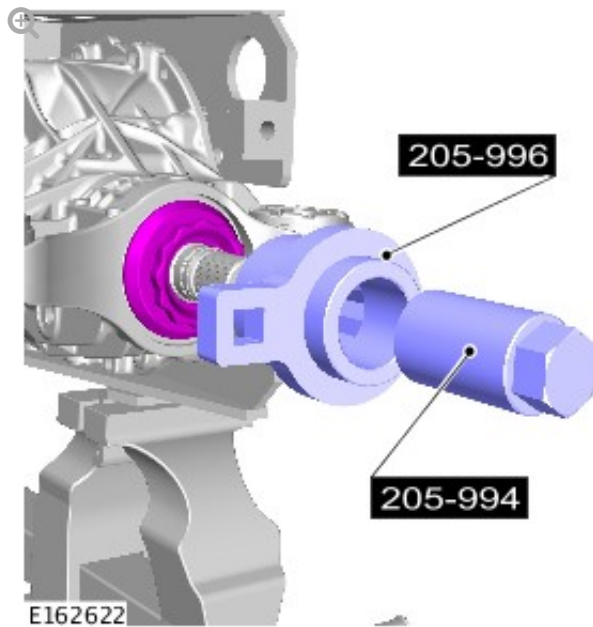


Install the Special Tool(s): [205-067](#), [JLR-205-994](#)

13.

CAUTION:

The special tool JLR-205-994 should be rotated clockwise to remove the drive pinion shaft nut.

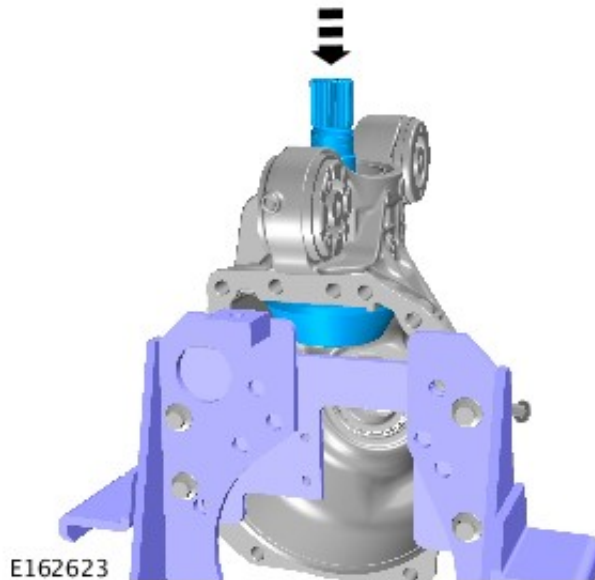


Install the Special Tool(s): [JLR-205-994](#), [JLR-205-996](#)

14.

CAUTIONS:

- Damage to the casing may occur if the base plates to support the special tool on the press are not used.
- Do not use a hammer to remove the drive pinion assembly. Failure to follow this instruction may cause damage to the component.



15.

WARNINGS:

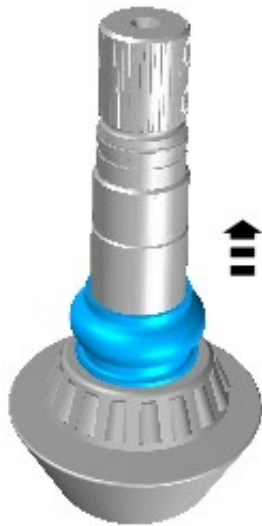
- Do not handle the head bearing.
- Do not clean or degrease the head bearing.

CAUTION:

Check the drive pinion assembly and bearing for any damage.

NOTES:

- Make sure to clean the threads prior to installation using Loctite cleaner 7200.
- Discard the drive pinion spacer.



E162344

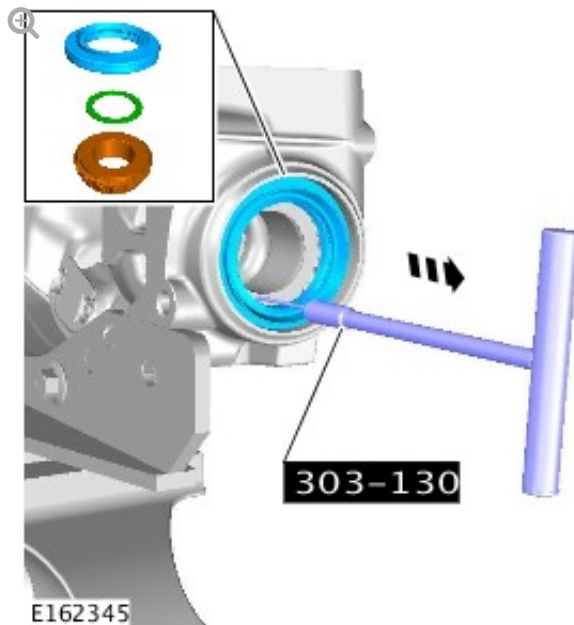
16.

CAUTION:

Make sure the drive pinion seal recess is not damaged as the seal is removed.

NOTE:

Discard the drive pinion seal, tail bearing thrust washer and tail bearing.

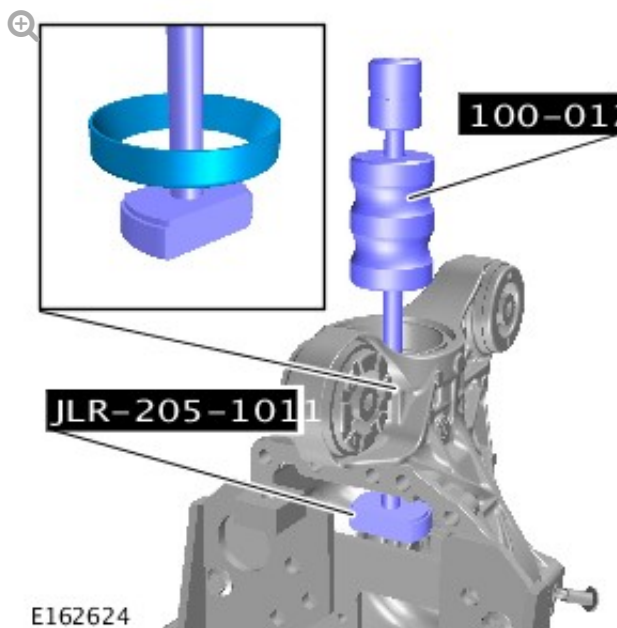


Install the Special Tool(s): [303-1300](#)

17.

WARNING:

Check the special tool for any damage before removing the drive pinion tail bearing outer race.



Using the special tools remove the drive pinion tail bearing outer race.

18.

CAUTION:

It is essential that absolute cleanliness is observed when working on the rear differential. Always cover any open orifices using lint free non-flocking material to prevent the ingress of foreign matter. Failure to follow this instruction may result damage to the components.

NOTE:

Inspect the drive pinion tail bearing outer race carrier for damage.

Using Loctite cleaner 7200 thoroughly clean the differential casings.

INSTALLATION

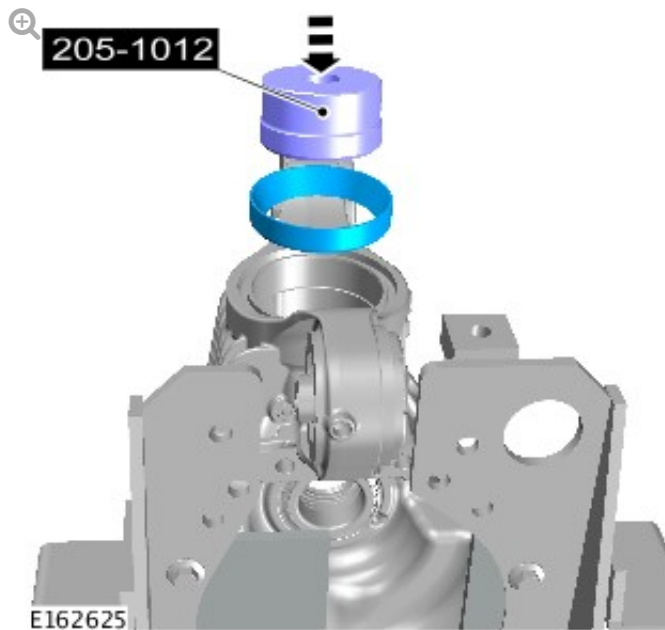
1.

WARNING:

Check the special tool for damage before installing the new drive pinion tail bearing outer race.

CAUTIONS:

- Do not use a hammer to install the new drive pinion tail bearing outer race. Failure to follow this instruction may cause damage to the component.
- Do not clean or lubricate the new drive pinion tail bearing, as it is supplied coated with a low friction oil. Failure to follow this instruction will require a new drive pinion tail bearing to be installed before the differential can be successfully assembled.



Using a suitable press, install the drive pinion tail bearing outer race.

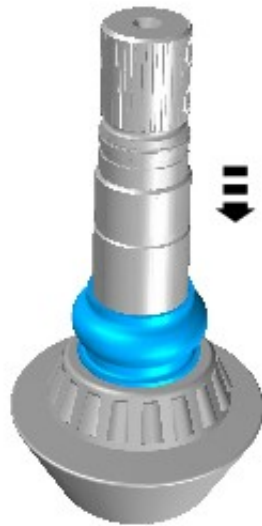
Special Tool(s): [JLR-205-1012](#)

Renew Part: [Differential pinion seal](#).

2.

NOTE:

Make sure to clean the threads prior to installation using Loctite cleaner 7200.

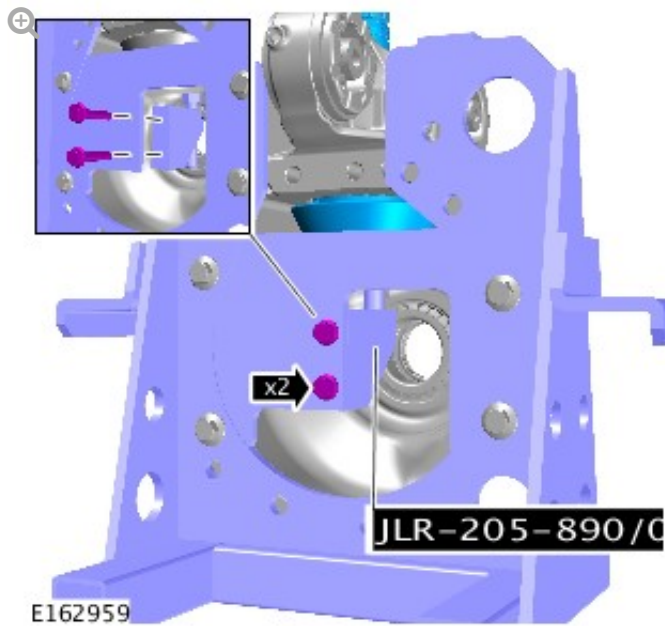


E162348

Install the new drive pinion spacer.

Renew Part: [Rear differential pinion repair kit](#).

3.



E162959

Torque: 45 Nm

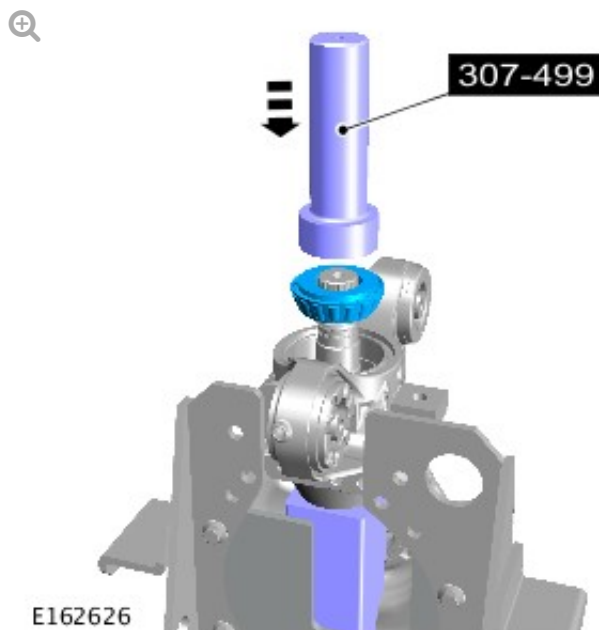
4.

WARNING:

Using the special tool, secure the drive pinion assembly.

CAUTIONS:

- Do not use a hammer to install the new drive pinion tail bearing. Failure to follow this instruction may cause damage to the component.
- Check the drive pinion assembly is secure before installing the drive pinion tail bearing.



Using a suitable press, install the new drive pinion tail bearing.

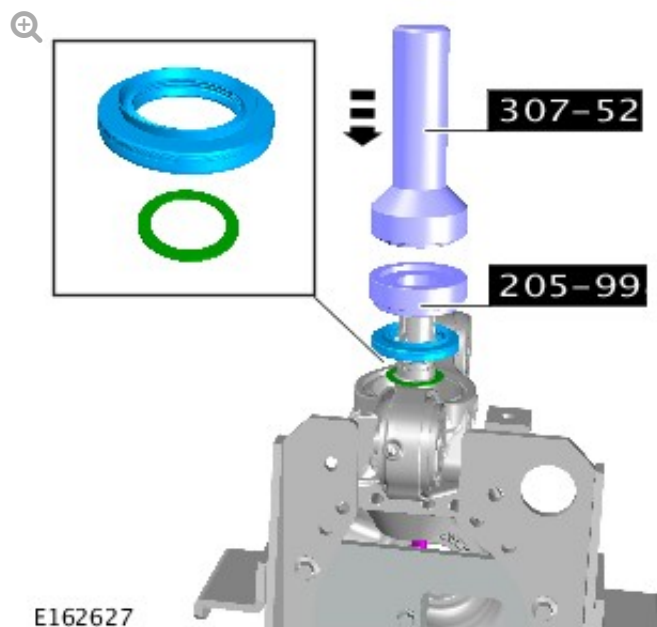
Special Tool(s): [307-499](#)

CAUTIONS:

- Check the special tool for damage before installing the new drive pinion seal.
- Check the special tool is clean and free from foreign material.
- Do not lubricate the drive pinion oil seal.
- Do not use a hammer to install the new drive pinion seal. Failure to follow this instruction may cause damage to the component.

NOTES:

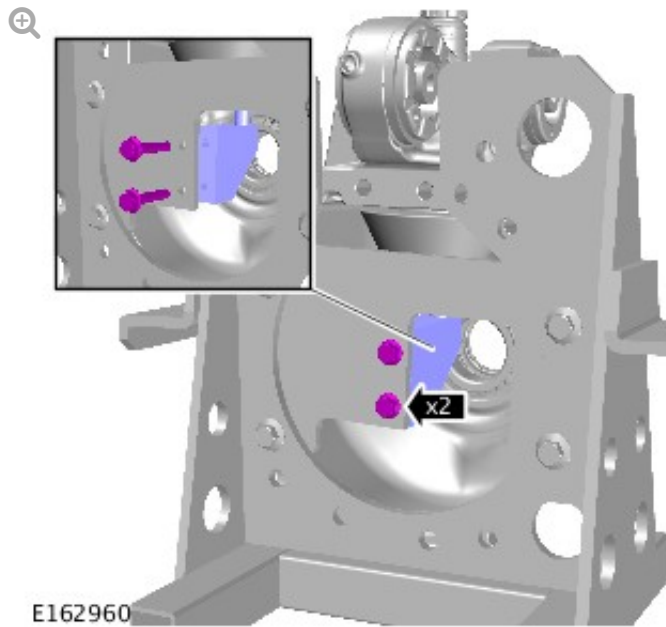
- Using the special tool, secure the drive pinion assembly.
- Install the new drive pinion tail bearing thrust washer.



Using a suitable press, install the new drive pinion seal.

Install the Special Tool(s): [JLR-205-998](#), [307-520](#)

6.



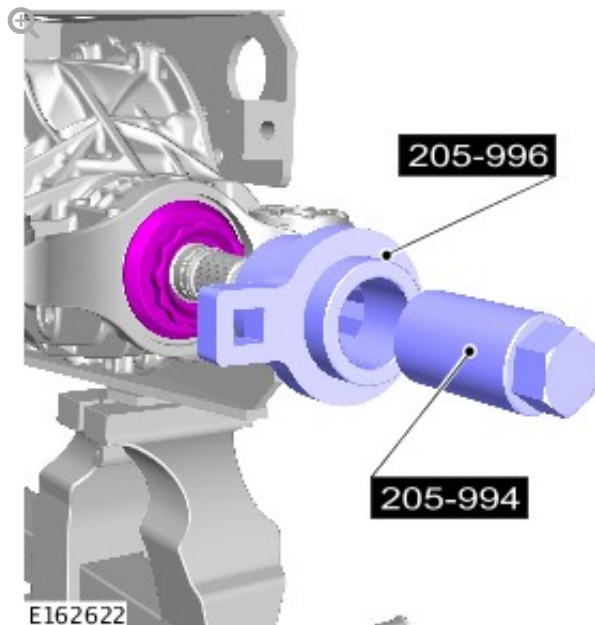
7.

WARNING:

High torque application.

CAUTIONS:

- The special tool JLR-205-994 should be rotated counter clockwise to install the new drive pinion nut.
- Do not use air tools to install the nut. Failure to follow this instruction may result in damage to the drive pinion.



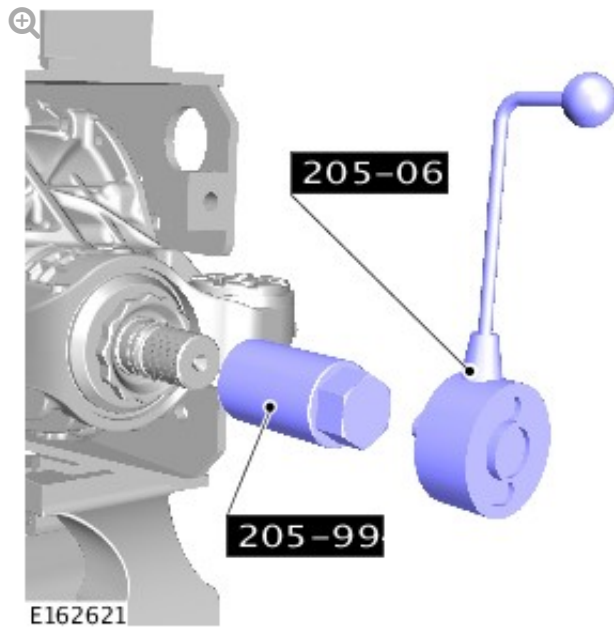
1. Apply Loctite 243 to the drive pinion threads.
2. Install the new drive pinion nut, tighten the nut until the collapsible spacer begins to collapse.

Install the Special Tool(s): [JLR-205-996](#), [JLR-205-994](#)

8.

CAUTIONS:

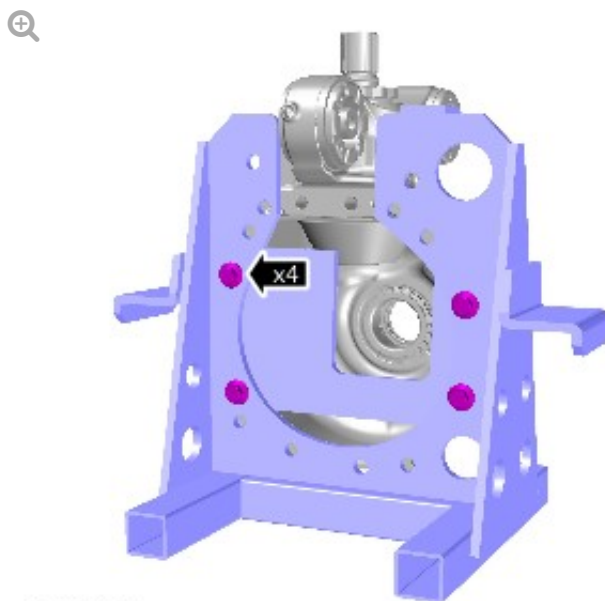
- Make sure the specified torque to rotate the drive pinion is not exceeded. If excess preload is applied to the joint the drive pinion should be removed and a new collapsible spacer, drive pinion tail bearing, bearing cup and flange nut must be installed.
- The drive pinion should be rotated through two full revolutions by hand before the torque measurement is performed.
- The special tool should be rotated at a constant steady speed to produce a consistent reading.
- The special tool 205-067 should be rotated clockwise to check the torque to rotate.



1. Check the torque required to rotate the drive pinion nut.
The specified torque is the torque to rotate figure recorded in step 9 of the removal process +0.45Nm
2. If the specified torque to rotate the drive pinion is not reached, tighten the pinion flange nut in one degree increments and check the rotational torque after each until the specified torque is achieved.

Install the Special Tool(s): [205-067](#), [JLR-205-994](#)

9.



10.



E162628

Install the crown wheel assembly.

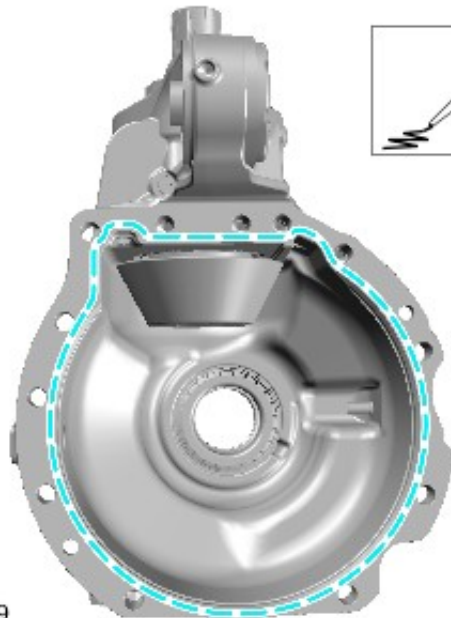
11.

WARNING:

Clean the bolt holes to remove any sealant and make sure the bolts threads are clean.

CAUTION:

Using Loctite cleaner 7200 thoroughly clean the mating faces, before the sealant is applied.



E162629

Apply a 3 Mm bead of sealant STC50550 to the areas shown.

12.

CAUTION:

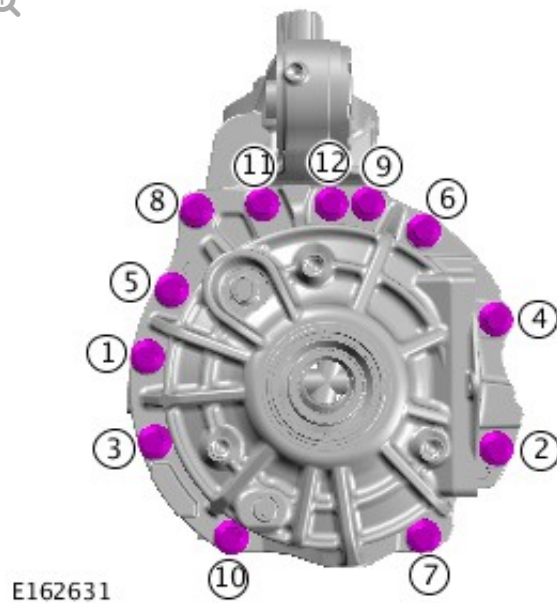
Vehicles with electronic rear differentials, make sure that the gear segment is aligned to allow correct installation of the locking pin.



E162630

Install the differential casing.

13.



Install the differential casing bolts in the sequence shown.

Torque: 120-125 Nm

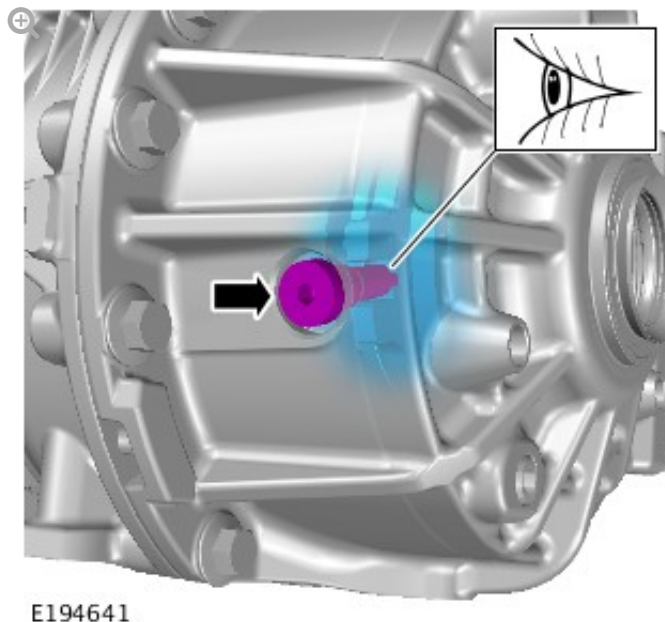
14.

CAUTION:

Make sure that the gear segment is aligned to allow correct installation of the locking pin.

NOTE:

If equipped.



E194641

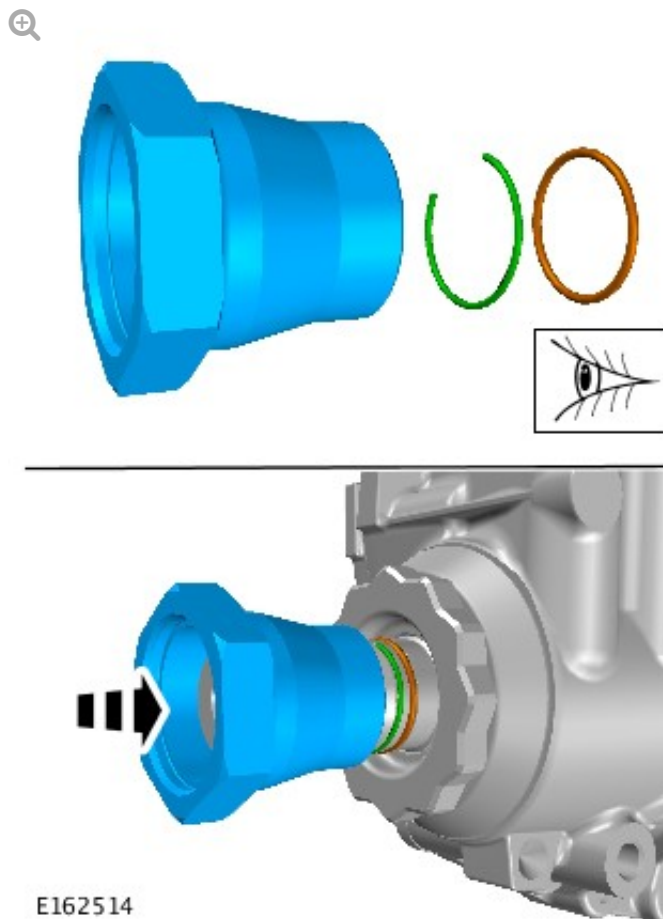
Install the electronic differential locking pin.

Torque: 22 Nm

15.

NOTES:

- Install the drive pinion o-ring seals.
- Install the drive pinion circlip.



Install the drive shaft nut.

Renew Part: [Rear differential pinion repair kit](#).

Renew Part: [Circlip](#).

16.

NOTE:

If equipped.

Refer to: [Differential Locking Motor](#) (205-02 Rear Drive Axle/Differential, Removal and Installation).

17. Refer to: [Differential Case - TDV6 3.0L Diesel /TDV8 4.4L Diesel/V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol](#) (205-02 Rear Drive Axle/Differential, Removal and Installation).

18. Refer to: [Differential Draining and Filling](#) (205-02 Rear Drive Axle/Differential, General Procedures).

